DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING

Experiment-5

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Subject Name: System Design
Subject Code: 23CSH-314

1. Aim: Design a Messenger Application(Similar to WhatsApp / Facebook Messenger): To design and analyze the architecture of a scalable real-time messenger application that supports one-to-one and group communication with high availability and low latency.

2. Objective:

- To implement real-time message delivery using WebSockets.
- To support one-to-one and group messaging.
- To preserve chat history.
- To ensure high availability and reliability.
- To design APIs for user management and chat operations.
- To design the system for large-scale usage (up to 1 Billion users).

3. Tools Used:

- Frontend (ReactJS, HTML, CSS, JavaScript, WebSocket) o Built responsive UI with real-time communication support
- Backend (Node.js / Java / Spring Boot) o Developed scalable REST APIs & WebSocket server
- Database (MongoDB / MySQL / PostgreSQL) o Managed structured & unstructured data efficiently
- Redis o Implemented caching & pub-sub for high performance
- Deployment (AWS / GCP, Load Balancer, Docker) o Cloud deployment with scalability, containerization & traffic distribution

4. System Requirements:

A. Functional Requirements

- User Registration
  a. POST /user/register
  b. Fields: userName, email, phoneNumber

- User Login
  a. POST /user/login
- One-to-One Messaging
  a. WS: /messages/send

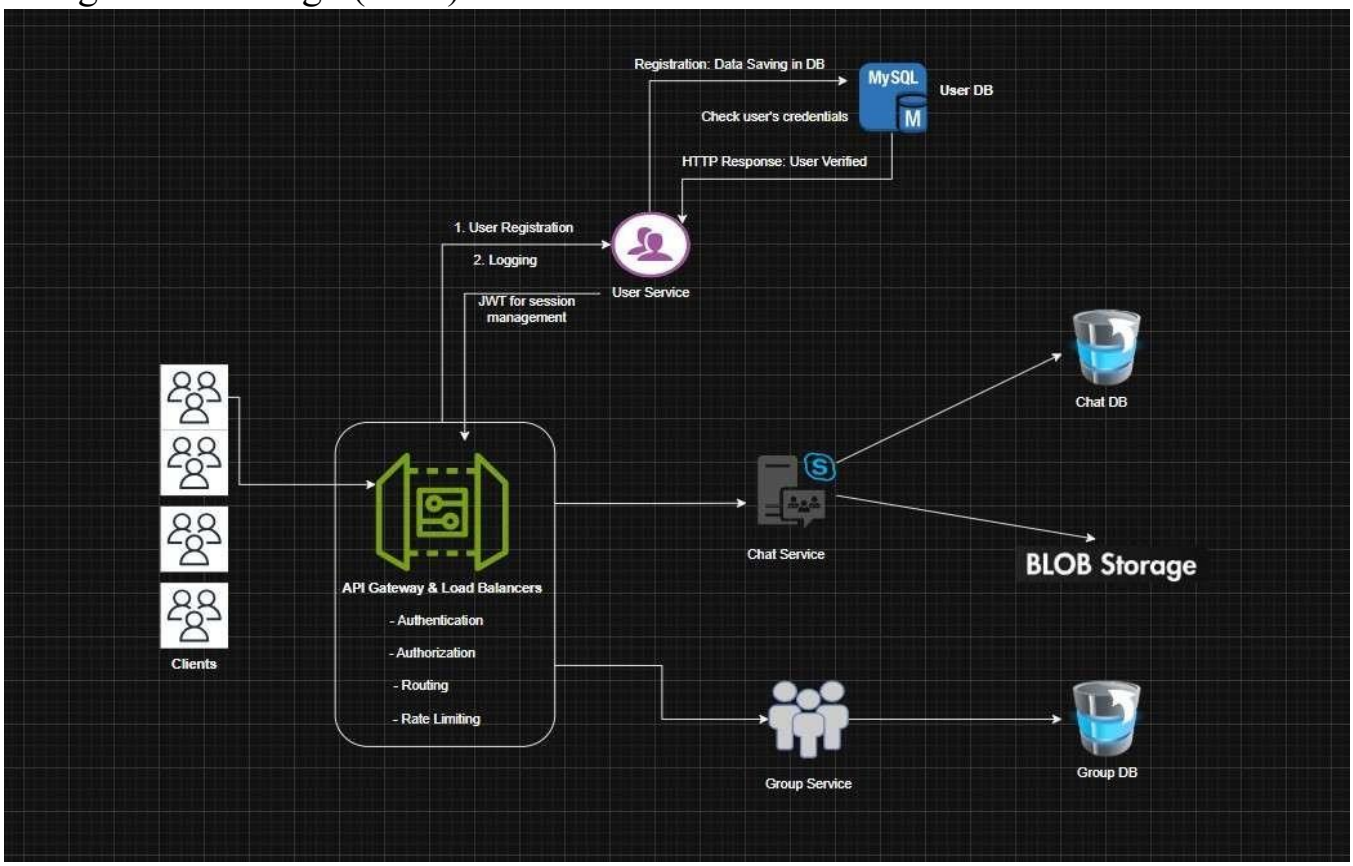
  b. GET /messages/{userId}/{receiverId}
  c. Pagination support
- Group Messaging
  a. POST /groups/create
  b. POST /groups/{groupId}/add
  c. DELETE /groups/{groupId}/remove
  d. WS: /messages/send
- Message Types
  a. Text messages
  b. Media messages (images, videos, files)
- Message History
  a. Persistent storage
  b. Chat list view
- Read Receipts
  a. Delivered
  b. Seen

B. Non-Functional Requirements

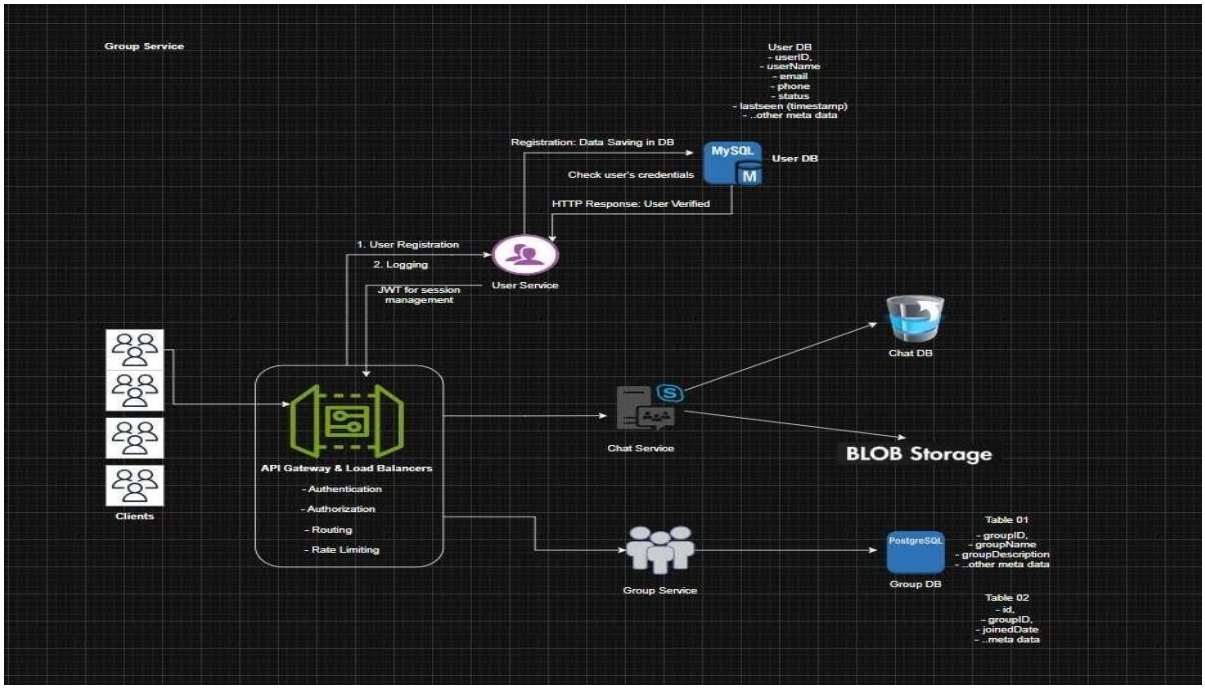
- Scalability o Target: 1 Billion users o 100 messages per user per day o ≈ 100 Billion messages/day o Estimated storage: ~100 TB

- CAP Theorem
  System prioritizes: o
  High Availability o
  Partition Tolerance o
  Eventual Consistency
- Latency: 200-300ms o Target message delivery time: 200–300 ms

5. High Level Design (HLD)



6. Low Level Design (LLD):



7. Learning Outcomes (What I Have Learnt)

- Understand real-time communication using WebSockets.
- Learn how to design scalable distributed systems.
- Apply CAP Theorem concepts in real-world system design.
- Design RESTful APIs and WebSocket endpoints.
- Analyze trade-offs between consistency, availability, and latency.

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